

Functions and transformations - Set 1 - Solutions

Specification MM1.7, MM8.2, MM8.3, MM8.4.

Answer key

Question	Answer	Topic
1	A	Functions and transformations
2	A	Functions and transformations
3	C	Functions and transformations
4	B	Functions and transformations
5	A	Functions and transformations
6	C	Functions and transformations
7	D	Functions and transformations
8	D	Functions and transformations
9	A	Functions and transformations
10	D	Functions and transformations
11	E	Functions and transformations
12	B	Functions and transformations

Worked solutions

Question 1 FUN-01 · Functions and transformations · spec MM8.2

The graph of $y = x^2$ is translated by 3 units in the positive x -direction and then stretched parallel to the y -axis with scale factor 2. What is the equation of the resulting graph?

Answer: A

Translation: $y = (x - 3)^2$. Stretch by factor 2 in the y -direction: $y = 2(x - 3)^2$.

Question 2 FUN-02 · Functions and transformations · spec MM8.2

$f(x) = x^2 - 4x$. The graph of $y = f(x + 2)$ has its minimum point at

Answer: A

$f(x) = (x - 2)^2 - 4$ has minimum at $(2, -4)$. Replacing x by $x + 2$ translates the graph 2 units to the left: $(0, -4)$.

Question 3 FUN-03 · Functions and transformations · spec MM8.2

Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $f(g(2)) - g(f(2))$.

Answer: C

$g(2) = 1$, $f(1) = 3$. $f(2) = 5$, $g(5) = 22$. So $3 - 22 = -19$.

Question 4 FUN-05 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ passes through $(4, 6)$. Which point must lie on the graph of $y = 3f(2x) - 1$?

Answer: B

$f(2x)$ halves x -coordinates: $(2, 6)$. Then $3f - 1$: $y = 18 - 1 = 17$. Point $(2, 17)$.

Question 5 FUN-06 · Functions and transformations · spec MM8.2

The curve $y = x^2 - 6x + 5$ is reflected in the y -axis. Find the equation of the reflected curve.

Answer: A

Reflecting in the y -axis replaces x with $-x$: $y = x^2 + 6x + 5$.

Question 6 FUN-08 · Functions and transformations · spec MM1.7

For which values of x is $\sqrt{x^2} = -x$?

Answer: C

$\sqrt{x^2} = |x|$, and $|x| = -x$ exactly when $x \leq 0$.

Question 7 FUN-09 · Functions and transformations · spec MM8.2

The graph of $y = \sin x$ is transformed to $y = 2\sin(3x) + 1$. What is the maximum value of the new function, and what is its period (in degrees)?

Answer: D

$\sin(3x)$ has period $360^\circ / 3 = 120^\circ$. $2\sin(3x)$ ranges from -2 to 2 , so $2\sin(3x) + 1$ has maximum 3.

Question 8 FUN-11 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ is transformed by the sequence: reflect in the x -axis, then translate 2 units in the positive y -direction, then reflect in the y -axis. The resulting graph has equation

Answer: D

Reflect in x -axis: $y = -f(x)$. Translate up 2: $y = -f(x) + 2$. Reflect in y -axis (replace x by $-x$): $y = 2 - f(-x)$.

Question 9 FUN-13 · Functions and transformations · spec MM8.4

The graph of $y = a(x + b)^2 + c$ has vertex $(-2, 5)$ and passes through $(0, -3)$. Find a .

Answer: A

$b = 2, c = 5$. At $x = 0$: $4a + 5 = -3$, so $a = -2$.

Question 10 FUN-14 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ has a maximum point at $(3, 8)$. The graph of $y = f(x - a) + b$ has its maximum at $(1, 2)$. Find $a + b$.

Answer: D

The maximum moves from $(3, 8)$ to $(3 + a, 8 + b) = (1, 2)$, so $a = -2, b = -6$ and $a + b = -8$.

Question 11 FUN-15 · Functions and transformations · spec MM8.2

Which of the following is the equation of the graph obtained by stretching $y = x^3 - 3x$ parallel to the x -axis with scale factor $\frac{1}{2}$?

Answer: E

A stretch parallel to the x -axis with factor $\frac{1}{2}$ gives $y = f(2x) = (2x)^3 - 3(2x) = 8x^3 - 6x$.

Question 12 FUN-17 · Functions and transformations · spec MM8.3

The line $y = mx + c$ passes through $(2, 7)$. If m is increased by 1 and c is decreased by 2, the new line passes through $(2, k)$. Find k .

Answer: B

New value at $x = 2$: $(m + 1) \cdot 2 + (c - 2) = 2m + c = 7$. So $k = 7$.